

Time Series

Applied Time Series Analysis **STA457 Lecture Notes**

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Time Series Analysis

Introduction

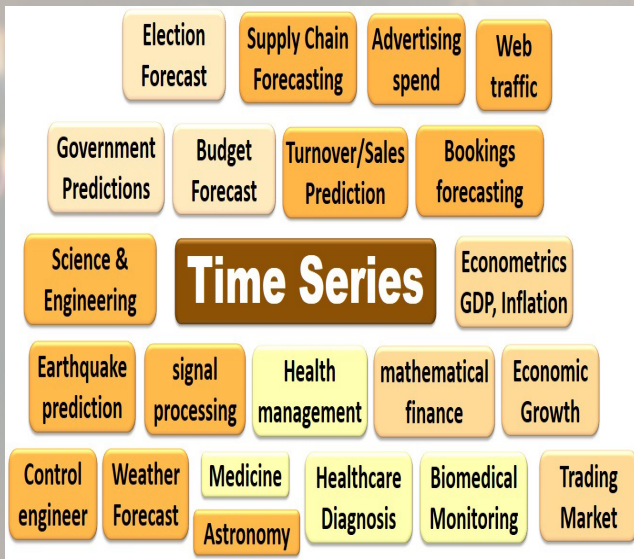


Figure 1: Real-world applications of time series.

Approaches Towards Time Series Analysis

- ➊ There exist two approaches towards analysis of time series data:
 - ➊ *Time Domain Approach:* The time domain approach views the investigation of *lagged* relationships as most important (e.g., how does what happened today affect what will happen tomorrow).
 - ➋ *Frequency Domain Approach:* Frequency domain approach views the investigation of *cycles* as most important (e.g., what is the economic cycle through periods of expansion and recession).
- ➋ Model-Building Strategy:
 - ➊ Model Specification
 - ➋ Model Fitting
 - ➌ Model Diagnostics

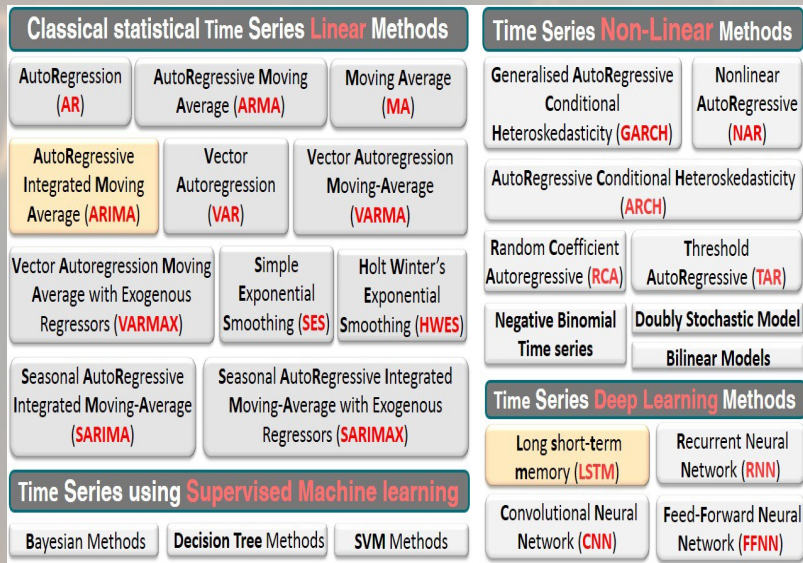


Figure 2: Different models for analysis of time series in time domain.

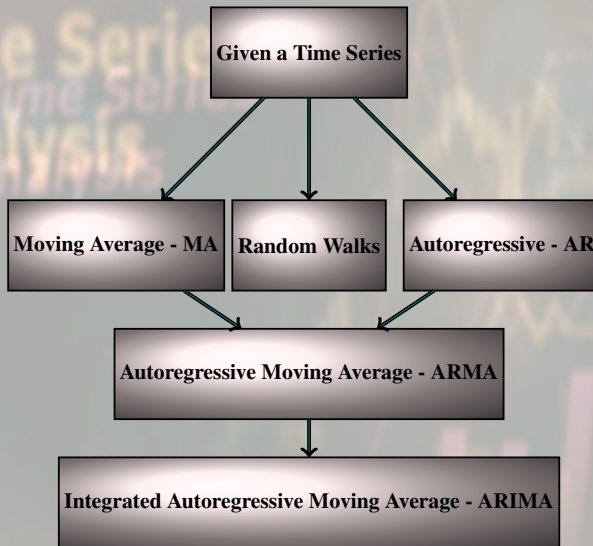


Figure 3: Some Linear Time Series Models.

Time Series Analysis

Basic Definitions

Random Variables and Probability Distributions

- ➊ A random variable is a *function* from sample space to the field of real numbers. A random variable is said to be discrete if its associated sample space is comprised of *discrete points*, whereas the sufficient condition for a random variable to be continuous is that its cumulative distribution function (CDF) needs to be *absolutely continuous* (i.e., with no discontinuity points).
- ➋ Random variables are denoted by uppercase Roman letters (e.g., X), while their *observed* values are represented by lowercase Roman letters (e.g., x). *Note that the text book does not follow this convention and denotes both the random variables and their observed values by uppercase letters.*
- ➌ The continuous random variable Z is said to follow the *normal distribution*, denoted by $Z \sim N(\mu, \sigma^2)$, if it has the following probability density function (PDF):

$$f_Z(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}. \quad (1)$$

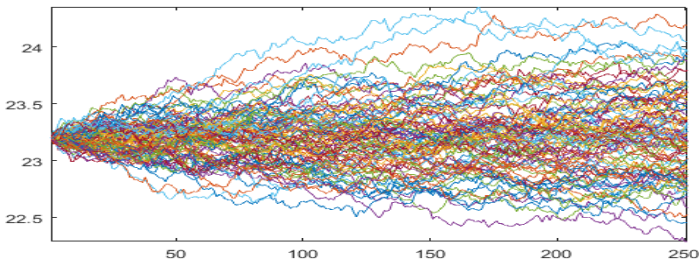
- ➍ A collection of random variables $\{X_1, X_2, \dots, X_n\}$ is *independent and identically distributed* (iid), if the random variables are *pairwise independent* and all of them share the *same underlying distribution* (i.e., they all have a common CDF).
Note that the given collection (set) of random variables is not necessarily ordered.

Time Series and Stochastic Processes

- 1 A collection of random variables $\{X_1, X_2, \dots, X_n\}$ forms a *stochastic process* (also called a *random process*) if its elements are *indexed over time*.
- 2 Note that the given collection (set) of random variables forming the stochastic process are necessarily **ordered**, and they should all take their values from a **common sample space**. However, the given elements of the sequence $\{X_t\}_{t=1}^n$ are not required to be iid in general (we use the subscript t to emphasize that the random variables $X_1, X_2, \dots, X_t, \dots, X_n$ are sequentially ordered (indexed) over time). It is also common to denote the time t random variable X_t by $X(t)$.
- 3 A *time series*, denoted by $\{x_t\}_{t=1}^n$ or $\{x_1, x_2, \dots, x_t, \dots, x_n\}$, is a collection of *observations* made *sequentially* through time from a *stochastic process* $\{X_t\}_{t=1}^n$.
- 4 A *stochastic process* can be described as a statistical phenomenon that evolves in time according to probabilistic laws, i.e., it does not deal with observations (**probabilistic approach**).
- 5 The goal of the *time series analysis* is to employ the observations made from an unknown stochastic process in order to make inference on the probabilistic laws that govern the underlying stochastic process (**statistical approach**).

What Makes the Analysis of Time Series More Challenging?

- 1 Each one of the random variables $X_1, X_2, \dots, X_{250}$ can take values according to their underlying probability distribution functions. So, there exist infinite number of potentially observed *paths* or *trajectories*. This infinite set of *time series* is sometimes called the *ensemble*, and every member of the ensemble, say the observed time series $x_1, x_2, \dots, x_{250}$, is referred to as a *realization* of the stochastic process.
- 2 Unlike most statistical problems which are concerned with estimating the properties of a population from a sample (of size greater than 1), in time series analysis there exist only a single outcome of the process (or a single realization). Time series analysis is then essentially concerned with evaluating the properties of the underlying probability model from this observed time series, even though this single realization is the only one we will ever observe.



Covariance Function

- ❶ Since most of our familiar summary statistics are not applicable to analysis of time series data, we introduce a new set of tools that make use of the *autocovariance* and *autocorrelation* functions. However, before that, it is extremely important for a time series analyst to be able to deal with and interpret the subtle concepts of *covariance function* and *correlation coefficient*.
- ❷ Recall that *variance* describes the *variability* of a random variable. In the same vein, *covariance* is a measure of how 2 random variables vary together. Or, the degree to which the *change* in one variable is *associated* with a *change* in another variable can mathematically be described in terms of the covariance of the variables.
- ❸ Suppose 2 random variables X and Y have means $\mathbb{E}[X] = \mu_X$ and $\mathbb{E}[Y] = \mu_Y$. Then, we define the covariance of X and Y as follows:

$$\gamma_{XY} := \text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (2)$$

- ❹ If X and Y are independent, then $\text{Cov}(X, Y) = 0$.
- ❺ Unlike variance which is always positive, covariance can take *negative* values:
 - ❶ $\text{Cov}(X, Y) > 0$ if *high* values of X tend to go with *high* values of Y .
 - ❷ $\text{Cov}(X, Y) < 0$ if *high* values of X tend to go with *low* values of Y .
- ❻ Covariance is a useful quantity for many mathematical purposes, but it is difficult to interpret, as it depends on the units in which X and Y are measured.

Correlation Coefficient

- ➊ One useful solution is to standardize the covariance between 2 random variables, which yields the *correlation coefficient*

$$\rho_{XY} = \frac{\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad (3)$$

where σ_X and σ_Y denote standard deviations of X and Y , respectively.

- ➋ Correlation coefficient lies within $-1 \leq \rho_{XY} \leq 1$ and is a useful measure of **linear association** between 2 random variables:

- ➌ $\rho_{XY} = 1$ if $Y = aX + b$ with $a > 0$
- ➍ $\rho_{XY} = -1$ if $Y = aX + b$ with $a < 0$

- ➎ Given N pairs of observations $\{(x_i, y_i) : i = 1, 2, \dots, N\}$, the usual estimate of ρ_{XY} is given by the **sample** correlation coefficient

$$r_{xy} = \frac{s_{xy}}{s_x s_y}, \quad (4)$$

where s_x^2 and s_y^2 are **sample** variances of X and Y , i.e., $s_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, and s_{xy} is an estimate of γ_{XY} given by

$$s_{xy} = \frac{1}{N - 1} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}). \quad (5)$$

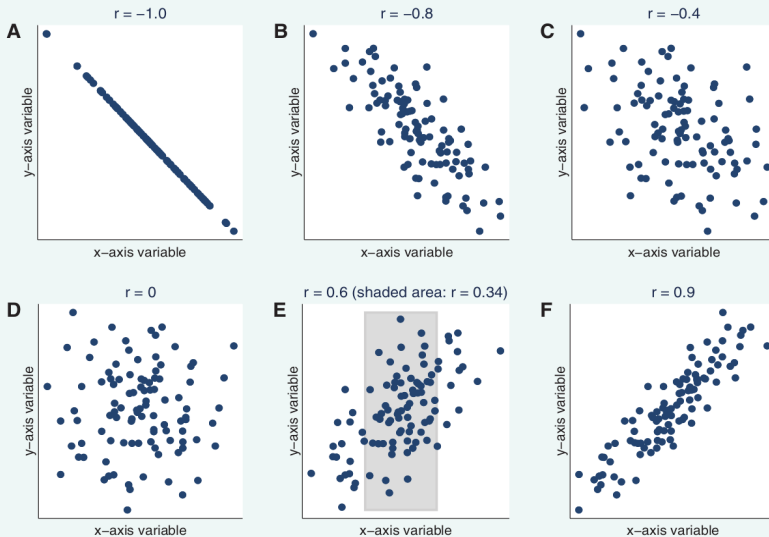


Figure 4: Adapted from (Schober, Boer, and Schwarte, 2018).

Autocovariance and Autocorrelation Coefficient Functions

- ❶ Let us assume a stochastic process $\{X_t : t = 0, \pm 1, \pm 2, \pm 3, \dots\}$ is given.
- ❷ Note that in this notation, it is assumed that the time origin (reference point) of the stochastic process (i.e., index $t = 0$) could potentially be shifted some units forward in time. Thus, the negative indexes would correspond to the random variables placed prior to X_0 and the positive indexes correspond to the random variables succeeding X_0 .
- ❸ For the given stochastic process, we define the *autocovariance function* $\gamma_{t,s}$ to be the covariance of X_t and X_s , namely

$$\gamma_{t,s} = \text{Cov}(X_t, X_s) = \mathbb{E}[(X_t - \mu_t)(X_s - \mu_s)] \quad t, s = 0, \pm 1, \pm 2, \pm 3, \dots \quad (6)$$

- ❹ Let us define a difference parameter $\tau = s - t$ and call it *lag*. Then, the autocovariance function can be expressed in terms of t and the lag parameter τ :

$$\gamma_{t,t+\tau} = \text{Cov}(X_t, X_{t+\tau}), \quad (7)$$

which is also referred to as the *autocovariance function at lag τ* . It is easy to see that $\gamma_{t,t+\tau}$ is a function of *both* t and τ .

- ❺ We can define an *autocorrelation coefficient function* as:

$$\rho_{t,s} = \frac{\text{Cov}(X_t, X_s)}{\sqrt{\text{Var}[X_t]\text{Var}[X_s]}} = \frac{\gamma_{t,s}}{\sqrt{\gamma_{t,t}\gamma_{s,s}}} \quad t, s = 0, \pm 1, \pm 2, \pm 3, \dots \quad (8)$$

Lemma (Properties of the Autocovariance Function)

- 1 $\gamma_{t,t} = \text{Var}[X_t]$
- 2 $\gamma_{t,s} = \gamma_{s,t}$
- 3 $|\gamma_{t,s}| \leq \sqrt{\gamma_{t,t}\gamma_{s,s}}$

Proof.

The proof for parts 1 and 2 are trivial.

To prove part 3, let α be a real variable, then

$$\text{Var}[\alpha X_t + X_s] \geq 0 \Rightarrow \text{Var}[X_t]\alpha^2 + 2\text{Cov}[X_t, X_s]\alpha + \text{Var}[X_s] \geq 0. \quad (9)$$

Now, we have a function of α that is a quadratic polynomial and is always nonnegative, so the discriminant should be nonpositive, i.e.,

$$(2\text{Cov}[X_t, X_s])^2 - 4\text{Var}[X_t]\text{Var}[X_s] \leq 0 \Rightarrow \text{Cov}^2[X_t, X_s] \leq \text{Var}[X_t]\text{Var}[X_s] \Rightarrow |\gamma_{t,s}| \leq \sqrt{\gamma_{t,t}\gamma_{s,s}}. \quad (10)$$



Lemma (Properties of the Autocorrelation Coefficient Function)

- 1 $\rho_{t,t} = 1$
- 2 $\rho_{t,s} = \rho_{s,t}$
- 3 $|\rho_{t,s}| \leq 1$

Time Series Analysis

References



Schober, Patrick, Christa Boer, and Lothar A Schwarte (2018). “Correlation coefficients: appropriate use and interpretation”. In: *Anesthesia & Analgesia* 126.5, pp. 1763–1768.